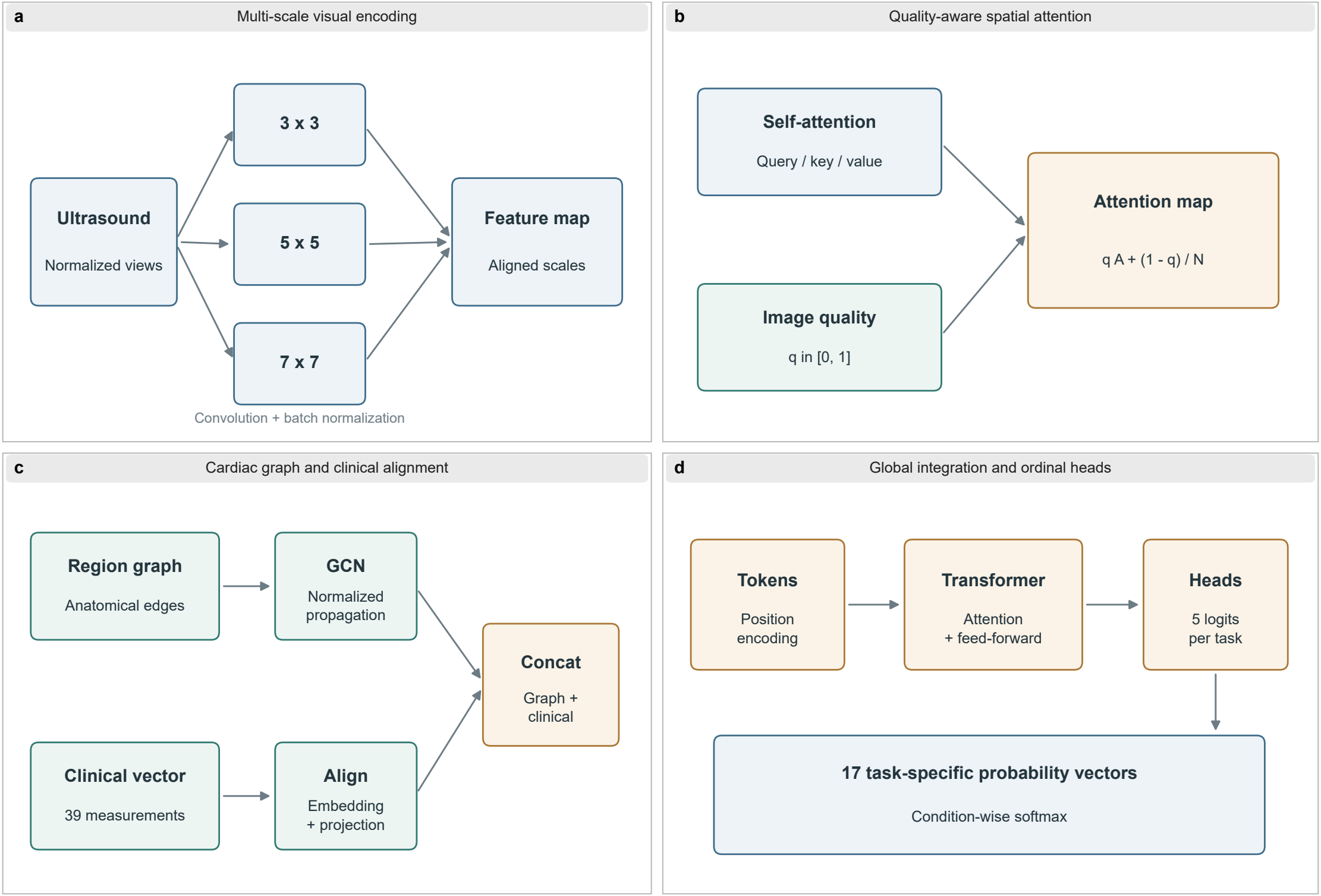
a
Multi-scale visual encoding
Ultrasound
Normalized views
3 x 3
5 x 5
7 x 7
Feature map
Aligned scales
Convolution + batch normalization
b
Quality-aware spatial attention
Self-attention
Query / key / value
Image quality
q in [0, 1]
Attention map
q A + (1 - q) / N
c
Cardiac graph and clinical alignment
Region graph
Anatomical edges
GCN
Normalized propagation
Clinical vector
39 measurements
Align
Embedding + projection
Concat
Graph + clinical
d
Global integration and ordinal heads
Tokens
Position encoding
Transformer
Attention + feed-forward
Heads
5 logits per task
17 task-specific probability vectors
Condition-wise softmax